# R vs East & nQuery

## Utilisation de R comme alternative aux logiciels propriétaires East et nQuery pour le calcul de Sample Size.

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vendredi 16 janvier 2026

Résumé Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis sagittis posuere ligula sit amet lacinia. Duis dignissim pellentesque magna, rhoncus congue sapien finibus mollis. Ut eu sem laoreet, vehicula ipsum in, convallis erat. Vestibulum magna sem, blandit pulvinar augue sit amet, auctor malesuada sapien. Nullam faucibus leo eget eros hendrerit, non laoreet ipsum lacinia. Curabitur cursus diam elit, non tempus ante volutpat a. Quisque hendrerit blandit purus non fringilla. Integer sit amet elit viverra ante dapibus semper. Vestibulum viverra rutrum enim, at luctus enim posuere eu. Orci varius natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus.

## Comparaison I : Sample size pour tester le hazard ratio dans un groupe à 2 bras.

### Méthode de Schoenfeld

Le calcul de la taille d'échantillon de Schoenfeld, comme les autres méthodes de log rank se fait sous les hypothèses de loi exponentielles et de proportional hazard qui ne sont pas complètement satisfaisantes dans certain cas mais complètement ancré dans l'analyse de survie en générale.

Prenons $\alpha$ l'erreur de type I et $\beta$ l'erreur de type II. Alors, en notant $e$ le nombre d'evènement nécessaire, la formule (simple) de Schoenfeld donne :

$$e = 2 \times \frac{2\left(z_{1-\alpha/2} + z_{1-\beta}\right)^2}{\log(\mathrm{HR})} \tag{1}$$

Où $z_x$ est le z-score de $x$, égale au quantile $(1-x)$ de la distribution normale standard.

### Méthode de Lakatos

blabla

## Paramètres

On va comparer les resultats de Rpact et de East sur l'ensemble de test suivant :

$\alpha \in \{0.01, 0.05, 0.1, 0.20, 0.49\}$

$\text{power} \in \{0.51, 0.8, 0.9, 0.99\}$

$\text{HR} \in \{0.1, 0.5, 0.7, 0.9, 0.99\}$

$\text{Survie à 3 ans} \in \{0.1, 0.3, 0.6, 0.9\}$

This is a block with gray background and slightly rounded corners.

## Rpact

On utilise la fonction `getSampleSizeSurvival` pour estimer le nombre d'évènements et de patients nécessaires.

```
Attaching package: 'rlang'
```

```
The following objects are masked from 'package:purrr':

    %@%, flatten, flatten_chr, flatten_dbl, flatten_int, flatten_lgl,
    flatten_raw, invoke, splice
```

## East

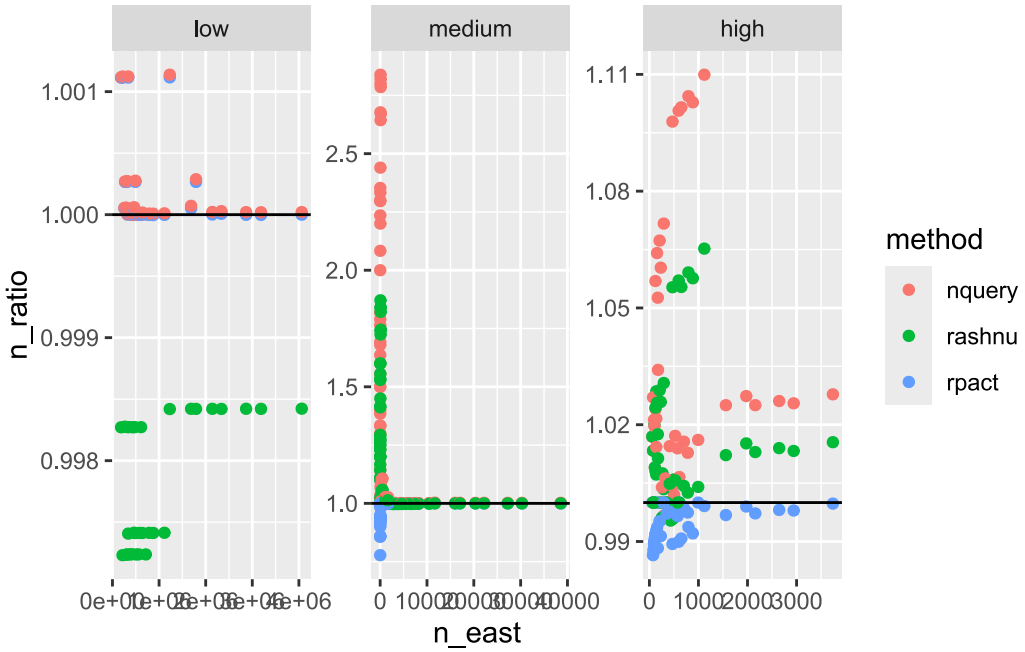
```
New names:
Rows: 76 Columns: 24
── Column specification ────────────────────────────────────────────────────────
Delimiter: "," chr
(4): ID, Design Type, Test Type, Var (Log HR) dbl (19): Sr. No., No. of
Looks,
Specified α, Power, nt/nc, Sample Size, Exp ... lgl (1): ... 24
ℹ Use `spec()` to retrieve the full column specification for this data. ℹ
Specify the column types or set `show_col_types = FALSE` to quiet this
message.
New names:
Rows: 96 Columns: 24
── Column specification ────────────────────────────────────────────────────────
Delimiter: "," chr
(4): ID, Design Type, Test Type, Var (Log HR) dbl (19): Sr. No., No. of
Looks,
Specified α, Power, nt/nc, Sample Size, Exp ... lgl (1): ... 24
ℹ Use `spec()` to retrieve the full column specification for this data. ℹ
Specify the column types or set `show_col_types = FALSE` to quiet this
message.
New names:
Rows: 96 Columns: 24
── Column specification
```

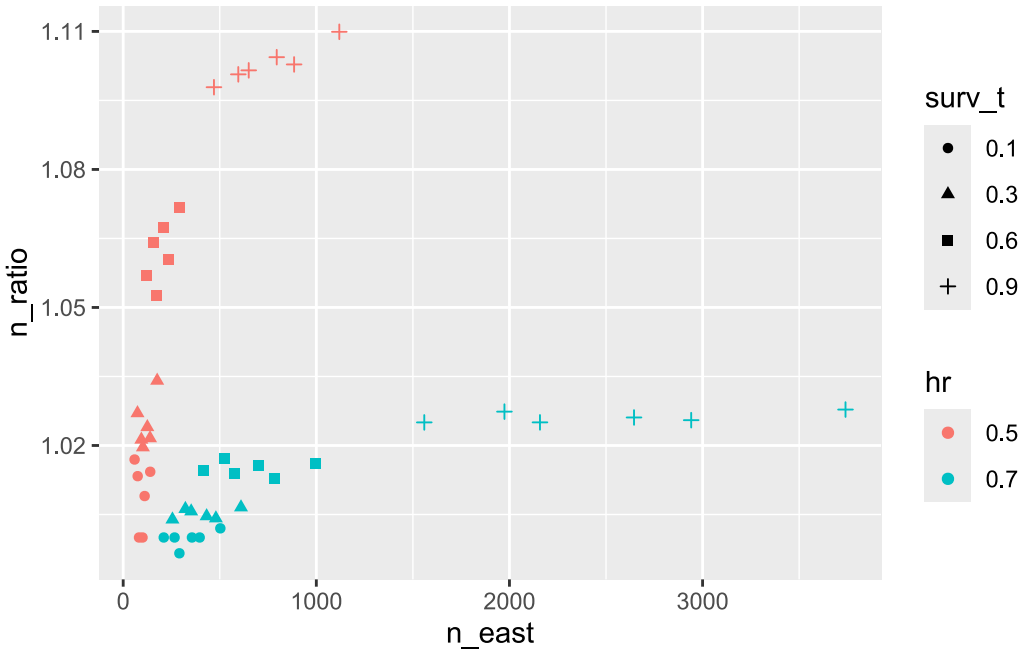
```
────────────────────────────────────────────────────────────────────────────────
Delimiter: "," chr
(4): ID, Design Type, Test Type, Var (Log HR) dbl (19): Sr. No., No. of
Looks,
Specified α, Power, nt/nc, Sample Size, Exp ... lgl (1):  ... 24
ℹ Use `spec()` to retrieve the full column specification for this data. ℹ
Specify the column types or set `show_col_types = FALSE` to quiet this
message.
New names:
Rows: 100 Columns: 24
── Column specification
────────────────────────────────────────────────────────────────────────────────
Delimiter: "," chr
(4): ID, Design Type, Test Type, Var (Log HR) dbl (19): Sr. No., No. of
Looks,
Specified α, Power, nt/nc, Sample Size, Exp ... lgl (1):  ... 24
ℹ Use `spec()` to retrieve the full column specification for this data. ℹ
Specify the column types or set `show_col_types = FALSE` to quiet this
message.
• `` → `...24`
```

## nQuery

## Rashnu

## Inter-method comparison






## Lakatos and Lan (1992)

### rpact

```
Attaching package: 'rlang'
```

```
The following objects are masked from 'package:purrr':

    %@%, flatten, flatten_chr, flatten_dbl, flatten_int, flatten_lgl,
    flatten_raw, invoke, splice
```

### East

```
New names:
Rows: 18 Columns: 24
── Column specification ────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────
Delimiter: "," chr
(4): ID, Design Type, Test Type, Var (Log HR) dbl (19): Sr. No., No. of
Looks,
Specified α, Power, nt/nc, Sample Size, Exp ... lgl (1): ... 24
ℹ Use `spec()` to retrieve the full column specification for this data. ℹ
Specify the column types or set `show_col_types = FALSE` to quiet this
message.
• `` → `...24`
```

nQuery

Rashnu

Comparision

| | | | n_rpact | n_east | nquery | rashnu | n_L | n_F | n_RGS | power_L | power_F | power_RGS |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0.8 | 0.667 | 1 | 1588 | 1593 | 1142 | 1620 | 1617 | 1628 | 1640 | 90.2 | 90.4 | 90.6 |
| | | 5 | 1979 | 1984 | 1414 | 2022 | 2017 | 2024 | 2046 | 90.6 | 90.7 | 91.0 |
| | | 9 | 2671 | 2679 | 1902 | 2728 | 2724 | 2709 | 2764 | 90.3 | 90.1 | 90.7 |
| | 0.500 | 1 | 601 | 604 | 356 | 638 | 638 | 649 | 664 | 90.1 | 90.6 | 91.2 |
| | | 5 | 749 | 753 | 440 | 798 | 798 | 807 | 831 | 90.7 | 91.0 | 91.8 |
| | | 9 | 1011 | 1017 | 590 | 1078 | 1079 | 1081 | 1124 | 90.1 | 90.2 | 91.2 |
| | 0.250 | 1 | 181 | 182 | 78 | 230 | 230 | 241 | 269 | 91.9 | 93.0 | 95.3 |
| | | 5 | 225 | 227 | 94 | 290 | 289 | 299 | 338 | 92.2 | 93.0 | 95.5 |
| | | 9 | 304 | 306 | 126 | 392 | 392 | 401 | 459 | 91.6 | 92.2 | 95.1 |
| 0.2 | 0.667 | 1 | 361 | 362 | 308 | 362 | 360 | 370 | 363 | 89.6 | 90.4 | 89.8 |
| | | 5 | 414 | 415 | 342 | 416 | 414 | 419 | 418 | 90.5 | 90.8 | 90.8 |
| | | 9 | 527 | 528 | 422 | 530 | 528 | 509 | 534 | 89.8 | 88.7 | 90.1 |
| | 0.500 | 1 | 133 | 134 | 102 | 134 | 134 | 144 | 138 | 89.7 | 91.7 | 90.5 |
| | | 5 | 154 | 155 | 112 | 156 | 156 | 164 | 161 | 89.9 | 91.3 | 90.8 |
| | | 9 | 196 | 197 | 138 | 200 | 200 | 200 | 207 | 89.7 | 89.7 | 90.7 |
| | 0.250 | 1 | 40 | 40 | 26 | 44 | 43 | 53 | 48 | 90.2 | 95.1 | 93.0 |
| | | 5 | 46 | 47 | 28 | 52 | 51 | 61 | 58 | 90.6 | 94.8 | 93.8 |
| | | 9 | 59 | 60 | 32 | 68 | 66 | 74 | 76 | 90.2 | 93.1 | 93.7 |

## Comparison II : Group sequential design

### Rpact

see rpact doc

### East

J'ai observé une variabilité entre 2 computes sur East de même paramètres, à vérifier.
Pas de variabilité sur rpact.

```
Rows: 81 Columns: 976
── Column specification
────────────────────────────────────────────────────────────────────────────────────────────────────
Delimiter: ","
chr   (3): GUID, DesignID, ProductVersion
```

```
dbl (961): TestID, ToCompute__A, WhatToComp__AA, DesignType__AAA,
TrialType_ ...
lgl  (12): Rfile__AACAA, Rfunc__AACAA, Rfile__AACAB, Rfunc__AACAB,
Rfile__AA ...

ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this
message.
```